The Language of Medicine 13th edition

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Chapter 13
Blood System

Chapter Goals (Slide 1 of 3)

Identify terms relating to the composition, formation, and function of blood.

Chapter Goals (Slide 2 of 3)

- Differentiate among the four major blood types.
- Differentiate among the different types of blood groups.
- Identify terms related to blood clotting.
- Build words and recognize combining forms used in blood system terminology.

Chapter Goals (Slide 3 of 3)

- Describe various pathologic conditions affecting blood.
- Describe various laboratory tests and clinical procedures used with hematologic disorders and recognize relevant abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 13
Lesson 13.1

Introduction

Blood: transports foods, gases, and wastes to and from the cells of the body

Other transported items:

- ➢Chemical messengers
- ➢Blood proteins, white blood cells, and platelets

Composition of Blood

- Cells
  - Erythrocytes
  - Leukocytes
  - Platelets
- Plasma
  - Water
  - Wastes
  - Nutrients
  - Salts
  - Hormones
  - Lipids
  - Vitamins

Cell Types and Function (Slide 1 of 2)

Cells—45% Blood Volume

Erythrocytes: red blood cells transport nutrients and oxygen

Leukocytes: white blood cells

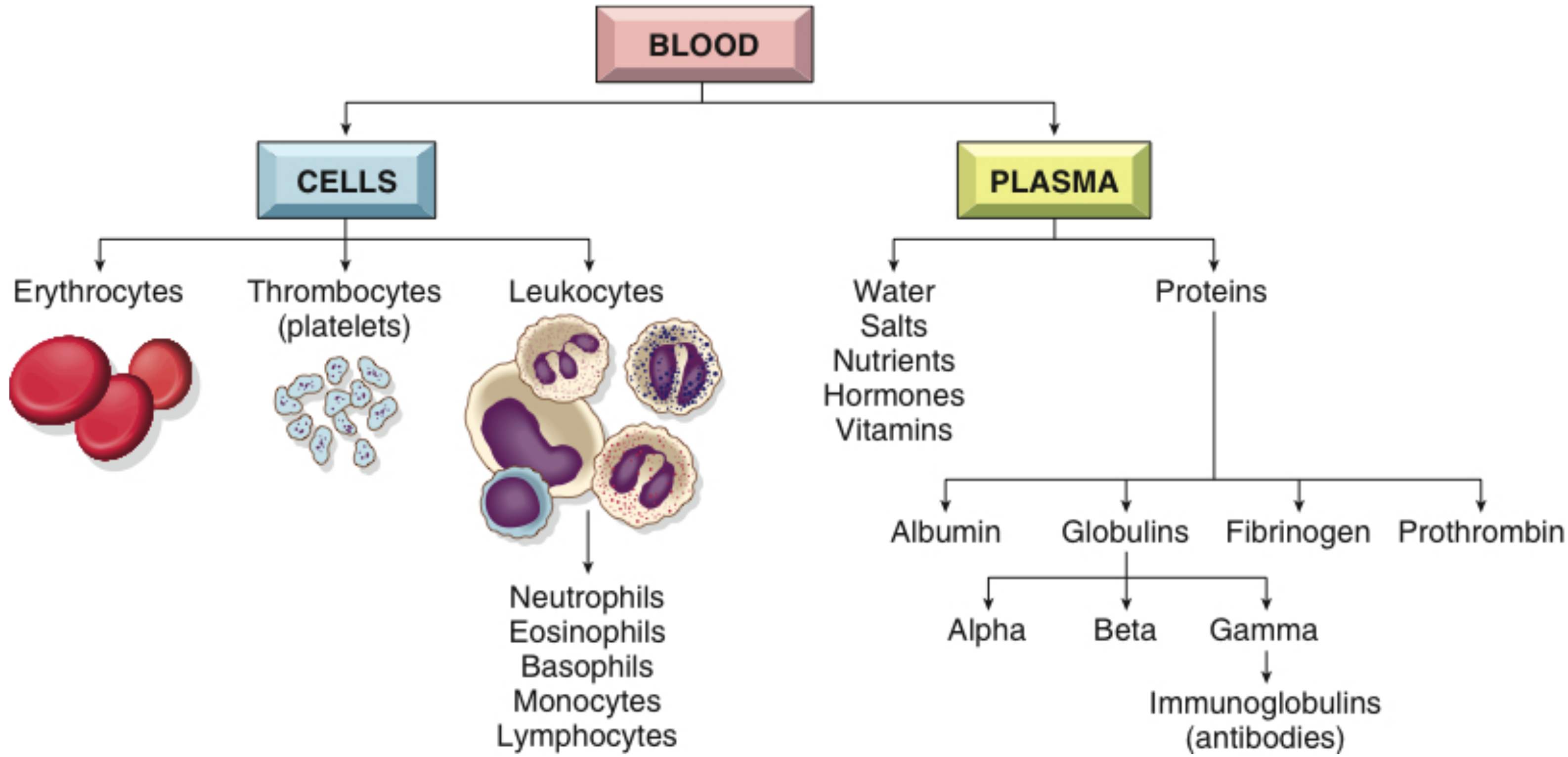
Thrombocytes: platelets; clot blood

Cell Types and Function (Slide 2 of 2)

Leukocytes: or white blood cells

- **Basophils:** contain heparin (prevents clotting) and histamine (involved in allergic responses)
- **Eosinophils:** phagocytic cells involved in allergic responses and parasitic infections
- **Neutrophils:** phagocytic cells that accumulate at sites of infection
- **Monocytes:** phagocytic cells that become macrophages and digest bacteria and tissue debris
- **Lymphocytes:** control the immune response; make antibodies to antigens

Review: Composition of Blood



Plasma

Plasma proteins

- ➢Albumin
- ➢Globulins: immunoglobulins (IgG, IgA, IgM, IgD, IgE)
- ➢Fibrinogen
- ➢Prothrombin

Blood Types

Type A: A antigen and anti-B antibody

Type B: B antigen and anti-A antibody

Type AB: A and B antigens and no antibodies (universal recipient)

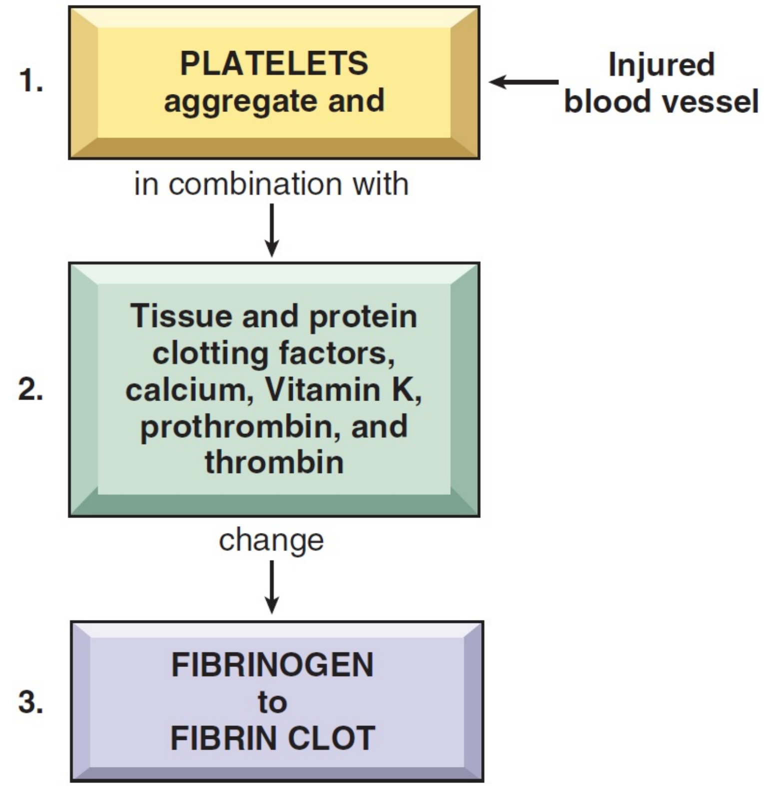
Type O: no A or B antigens and both anti-A and anti-B antibodies (universal donor)

Rh factor (positive and negative)

Blood Clotting

Coagulation: fibrin clot

Anticoagulants: heparin, warfarin (Coumadin)



QUICK QUIZ (Slide 1 of 3)

1. **The blood contains which of the following to transport oxygen?**

   A. White blood cells
   B. Plasma
   C. Platelets
   D. Red blood cells

QUICK QUIZ (slide 2 of 3)

2. Blood contains which clotting cells?

A. Erythrocytes
B. Plasma
C. Thrombocytes
D. Leukocytes

Vocabulary (Slide 1 of 28)

Term
albumin
antibody (Ab)
antigen

Vocabulary (Slide 2 of 28)

Term	Meaning/Definition
albumin	Protein in the blood; maintains the proper amount of water in blood
antibody (Ab)	A specific protein produced by the lymphocytes in response to bacteria, viruses, or other antigens
antigen	A substance that stimulates production of an antibody

Vocabulary (Slide 3 of 28)

Term
basophil
bilirubin
coagulation

Vocabulary (Slide 4 of 28)

Term	Meaning/Definition
basophil	White blood cell that contains granules that stain blue
bilirubin	Orange-yellow pigment in bile; formed by breakdown of hemoglobin when RBCs are destroyed
coagulation	Blood clotting

Vocabulary (Slide 5 of 28)

Term
colony-stimulating factor (CSF)
differentiation
electrophoresis

Vocabulary (Slide 6 of 28)

Term	Meaning/Definition
colony-stimulating factor (CSF)	Protein that stimulates growth of white blood cells
differentiation	The change in structure and function of a cell as it matures; specialization
electrophoresis	A method of separating serum proteins by electrical charge and size

Vocabulary (Slide 7 of 28)

Term
eosinophil
erythroblast
erythrocyte

Vocabulary (Slide 8 of 28)

Term	Meaning/Definition
eosinophil	White blood cell that contains granules that stain red
erythroblast	An immature red blood cell
erythrocyte	A red blood cell

Vocabulary (Slide 9 of 28)

Term
erythropoietin (EPO)
fibrin
fibrinogen

Vocabulary (Slide 10 of 28)

Term	Meaning/Definition
erythropoietin (EPO)	Hormone secreted by the kidneys that stimulates red blood cell formation
fibrin	Protein that forms the basis of a blood clot
fibrinogen	Plasma protein that is converted to fibrin in the clotting process

Vocabulary (Slide 11 of 28)

Term
globulin
granulocyte
hematopoietic stem cell

Vocabulary (Slide 12 of 28)

Term	Meaning/Definition
globulin	Plasma protein
granulocyte	White blood cell with numerous dark-staining granules
hematopoietic stem cell	A cell in bone marrow that gives rise to all types of blood cells

Vocabulary (Slide 13 of 28)

Term
hemoglobin
hemolysis
heparin

Vocabulary (Slide 14 of 28)

Term	Meaning/Definition
hemoglobin	Blood protein containing iron; carries oxygen in red blood cells
hemolysis	Breakdown of red blood cells
heparin	An anticoagulant found in blood and tissue cells

Vocabulary (Slide 15 of 28)

Term
immune reaction
immunoglobulin
leukocyte

Vocabulary (Slide 16 of 28)

Term	Meaning/Definition
immune reaction	Response of the immune system to foreign invasion
immunoglobulin	A protein with antibody activity
leukocyte	A white blood cell

Vocabulary (Slide 17 of 28)

Term
lymphocyte
macrophage
megakaryocyte

Vocabulary (Slide 18 of 28)

Term	Meaning/Definition
lymphocyte	Mononuclear leukocyte that produces antibodies
macrophage	Monocyte that migrates from the blood to tissue spaces; as a phagocyte, it engulfs foreign material and debris; destroys worn out RBCs
megakaryocyte	Large platelet precursor cell found in the bone marrow

Vocabulary (Slide 19 of 28)

Term
monocyte
mononuclear
myeloblast

Vocabulary (Slide 20 of 28)

Term	Meaning/Definition
monocyte	Leukocyte with one large nucleus; engulfs foreign material and debris; becomes macrophage
mononuclear	Pertaining to cell (leukocyte) with single round nucleus
myeloblast	Immature bone marrow that gives rise to granulocytes

Vocabulary (Slide 21 of 28)

Term
neutrophil
plasma
plasmapheresis

Vocabulary (Slide 22 of 28)

Term	Meaning/Definition
neutrophil	Granulocytic leukocyte formed in bone marrow; polymorphonuclear leukocyte
plasma	Liquid portion of blood; contains water, proteins, salts, nutrients, lipids, hormones, and vitamins
plasmapheresis	Removal of plasma from withdrawn blood by centrifuge

Vocabulary (Slide 23 of 28)

Term
platelet
polymorphonuclear
prothrombin

Vocabulary (Slide 24 of 28)

Term	Meaning/Definition
platelet	A small blood fragment important in clotting
polymorphonuclear	Pertaining to a white blood cell with multi-shaped nucleus; neutrophil
prothrombin	Plasma protein; converted to thrombin in the clotting process

Vocabulary (Slide 25 of 28)

Term
reticulocyte
Rh factor
serum

Vocabulary (Slide 26 of 28)

Term	Meaning/Definition
reticulocyte	Immature erythrocyte
Rh factor	Antigen on red blood cells of Rh-positive (RH^+) individuals
serum	Plasma minus clotting proteins and cells

Vocabulary (Slide 27 of 28)

Term
stem cell
thrombin
thrombocyte

Vocabulary (Slide 28 of 28)

Term	Meaning/Definition
stem cell	Unspecialized cell that gives rise to mature, specialized forms
thrombin	Enzyme that converts fibrinogen to fibrin during coagulation
thrombocyte	Platelets

Combining Forms and Terminology (Slide 1 of 4)

Combining Form	Meaning
bas/o	base
chrom/o	color
coagul/o	clotting
cyt/o	cell
eosin/o	red, dawn, rosy
erythr/o	red

Combining Forms and Terminology (Slide 2 of 4)

Combining Form	Meaning
granul/o	granules
hem/o	blood
hemat/o	blood
hemoglobin/o	hemoglobin
is/o	same, equal
kary/o	nucleus

Combining Forms and Terminology (Slide 3 of 4)

Combining Form	Meaning
leuk/o	white
mon/o	one, single
morph/o	shape, form
myel/o	bone marrow
neutr/o	neutral
nucle/o	nucleus

Combining Forms and Terminology (Slide 4 of 4)

Combining Form	Meaning
phag/o	eat, swallow
poikil/o	varied, irregular
sider/o	iron
spher/o	globe, round
thromb/o	clot

Suffixes

- -apheresis
- -blast
- -cyte
- -cytosis
- -emia
- -gen
- -globin
- -globulin
- -lytic
- -oid
- -osis
- -penia
- -phage
- -philia
- -phoresis
- -poiesis
- -stasis

Chapter 13
Lesson 13.2

Diseases of Red Blood Cells: Anemia

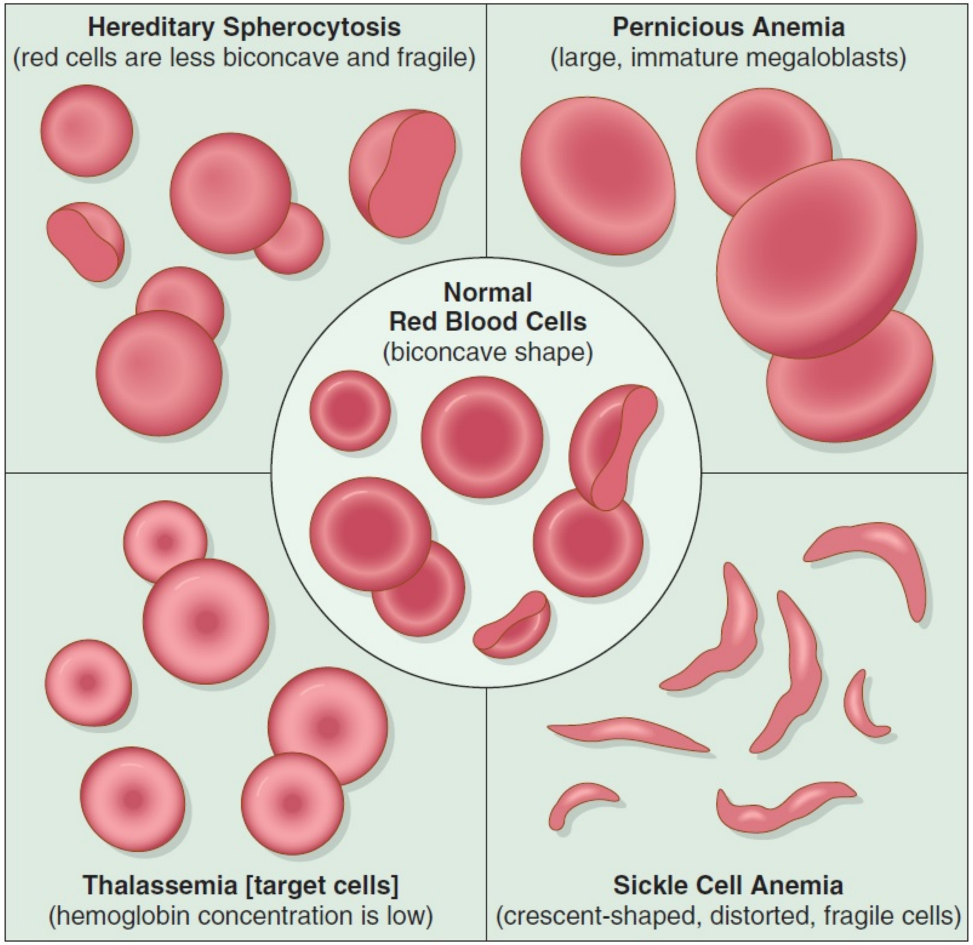
Anemia: A deficiency in erythrocytes or hemoglobin

- **Aplastic anemia**
- **Hemolytic anemia**
- **Pernicious anemia**
- **Sickle cell anemia**
- **Thalassemia**

Types of Anemia (Slide 1 of 2)

- **Aplastic anemia:** failure of blood cell production due to aplasia or absence of cell formation of bone marrow cells
- **Hemolytic anemia:** reduction in red cells due to excessive destruction
- **Pernicious anemia:** lack of mature erythrocytes caused by inability to absorb vitamin B_{12} into the bloodstream
- **Sickle cell:** hereditary disorder of abnormal hemoglobin producing sickle shape erythrocytes and hemolysis
- **Thalassemia:** an inherited defect in the ability to produce hemoglobin, leading to hypochromia

Types of Anemia (Slide 2 of 2)



QUICK QUIZ (Slide 3 of 3)

3. **What is the most common type of anemia?**

    A. Hemolytic anemia
    B. Sickle cell anemia
    C. Iron deficiency anemia
    D. Aplastic anemia

Other Diseases of Red Blood Cells

- **Hemochromatosis:** excess iron deposits throughout the body
- **Polycythemia vera:** general increase in red blood cells

Disorders of Blood Clotting

Hemophilia: excessive bleeding caused by hereditary lack of factors VIII and IX necessary for blood clotting. Patients often bleed into weight-bearing joints, especially the ankles and knees

Purpura: multiple pinpoint hemorrhages and accumulation of blood under the skin

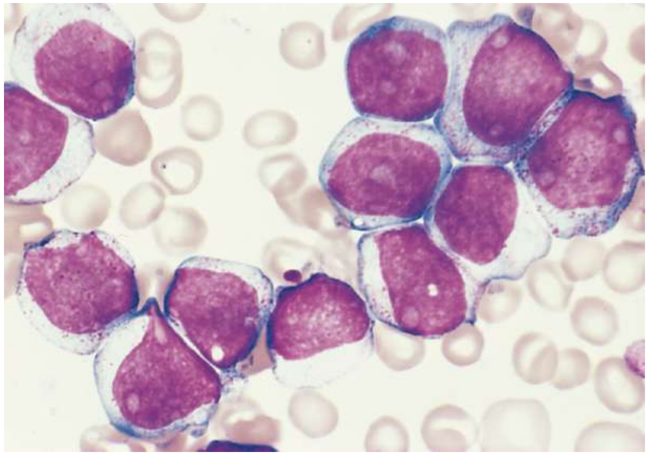
Diseases of White Blood Cells: Leukemia

Leukemia: an increase in cancerous white blood cells

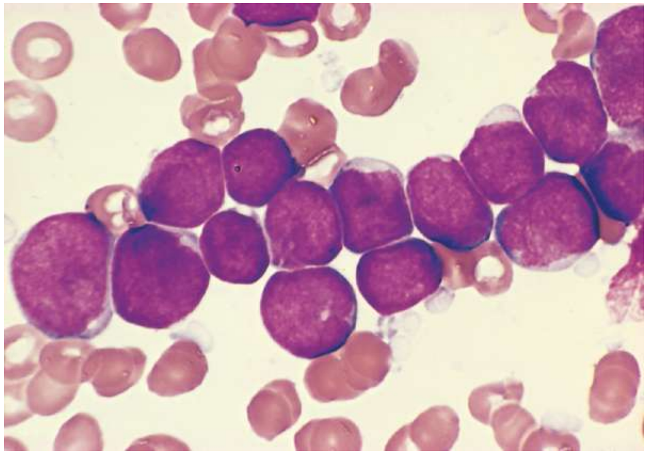
- ➢Acute myeloid leukemia (AML)
- ➢Acute lymphoid leukemia (ALL)
- ➢Chronic myeloid leukemia (CML)
- ➢Chronic lymphoid leukemia (CLL)

Acute Leukemia

Acute myeloid leukemia



Acute lymphoid leukemia



Other Diseases of White Blood Cells

Granulocytosis: abnormal increase in granulocytes in the blood

Mononucleosis: an infectious disease marked by increased numbers of mononuclear leukocytes and enlarged cervical lymph nodes

Disease of Bone Marrow Cells

Multiple myeloma: malignant neoplasm of bone marrow. Malignant cells (lymphocytes called plasma cells that produce antibodies) destroy bone tissue and cause overproduction of immunoglobulins, including Bence Jones protein.

Chapter 13
Lesson 13.3

Laboratory Blood Tests (Slide 1 of 2)

- Antiglobulin test (Coombs test)
- Complete blood count (CBC)
- Erythrocyte sedimentation rate (ESR)
- Hematocrit (Hct)
- Hemoglobin test (H, Hg, Hgb)
- Partial thromboplastin time (PTT)

Laboratory Blood Tests (Slide 2 of 2)

- Platelet count
- Prothrombin time (PT)
- Red blood cell count (RBC)
- Red blood cell morphology
- White blood cell count (WBC)
- White blood cell differential

Clinical Procedures

Apheresis: separation of blood into component parts and removal of a select part from the blood

Blood transfusion: whole blood or cells taken from a donor and infused into a patient

Bone marrow biopsy: microscopic examination of a core of bone marrow removed with a needle

Hematopoietic stem cell transplantation: peripheral stem cells from a compatible donor administered into a recipient

Abbreviations (Slide 1 of 5)

Abbreviation	Meaning
Ab	Antibody
ABMT	Autologous bone marrow transplantation
ABO	Four main blood types – A, B, AB, and O
ALL	Acute lymphoid leukemia
AML	Acute myeloid leukemia
ANC	Absolute neutrophil count
ASCT	Autologous stem cell transplantation
banda	Immature white blood cells (granulocytes)
baso	Basophils
BMT	Bone marrow transplantation

Abbreviations (Slide 2 of 5)

Abbreviation	Meaning
CBC	Complete blood count
CLL	Chronic lymphoid leukemia
CML	Chronic myeloid leukemia
DIC	Disseminated intravascular coagulation
diff	Differential count (white blood cells)
EBV	Epstein-Barr virus
eos	Eosinophils
EPO	Erythropoietin
ESR	Erythrocyte sedimentation rate
Fe	Iron

Abbreviations (Slide 3 of 5)

Abbreviation	Meaning
G-CSF	Granulocyte colony-stimulating factor
GM-CSF	Granulocyte-macrophage colony-stimulating factor
g/dL	Gram per deciliter
GVHD	Graft-versus-host disease
HCL	Hairy cell leukemia
Hct	Hematocrit
Hgh, HGB	Hemoglobin
H and H	Hemoglobin and hematocrit
HLA	Human leukocyte antigen

Abbreviations (Slide 4 of 5)

Abbreviation	Meaning
IgS, IgD, IgE, IgG, IgM	Immunoglobulins
ITP	Idiopathic thromocytic purpura
lymphs	Lymphocytes
MCH	Mean corpuscular hemoglobin
MCHC	Mean corpuscular hemoglobin concentration
MCV	Mean corpuscular volume
MDS	Myelodysplastic syndrome
Mm^3	Cubic millimeter
mono	Monocyte
Polys, PMNs, PMNLs	Polymorphonuclear leukocytes

Abbreviations (Slide 5 of 5)

Abbreviation	Meaning
PT, pro time	Prothrombin time
PTT	Partial thromboplastin time
RBC	Red blood cell; red blood cell c
sed rate	Erythrocyte sedimentation rate
segs	Segmented, mature white blood cells
SMAC	Sequential Multiple Analyzer Computer
WBC	White blood cell; white blood cell count
WNL	Within normal limits

Review Sheet – Combining Forms (Slide 1 of 8)

Combining Form
bas/o
chrom/o
coagul/o
cyt/o
eosin/o
erythr/o

Review Sheet – Combining Forms (Slide 2 of 8)

Combining Form	Meaning
bas/o	base
chrom/o	color
coagul/o	clotting
cyt/o	cell
eosin/o	red, dawn, rosy
erythr/o	red

Review Sheet – Combining Forms (Slide 3 of 8)

Combining Form
granul/o
hem/o
hemat/o
hemoglobin/o
is/o
kary/o

Review Sheet – Combining Forms (Slide 4 of 8)

Combining Form	Meaning
granul/o	granules
hem/o	blood
hemat/o	blood
hemoglobin/o	hemoglobin
is/o	same, equal
kary/o	nucleus

Review Sheet – Combining Forms (Slide 5 of 8)

Combining Form
leuk/o
mon/o
morph/o
myel/o
neutr/o
nucle/o

Review Sheet – Combining Forms (Slide 6 of 8)

Combining Form	Meaning
leuk/o	white
mon/o	one, single
morph/o	shape, form
myel/o	bone marrow
neutr/o	neutral
nucle/o	nucleus

Review Sheet – Combining Forms (Slide 7 of 8)

Combining Form
phag/o
poikil/o
sider/o
spher/o
thromb/o

Review Sheet – Combining Forms (Slide 8 of 8)

Combining Form	Meaning
phag/o	eat, swallow
poikil/o	varied, irregular
sider/o	iron
spher/o	globe, round
thromb/o	clot